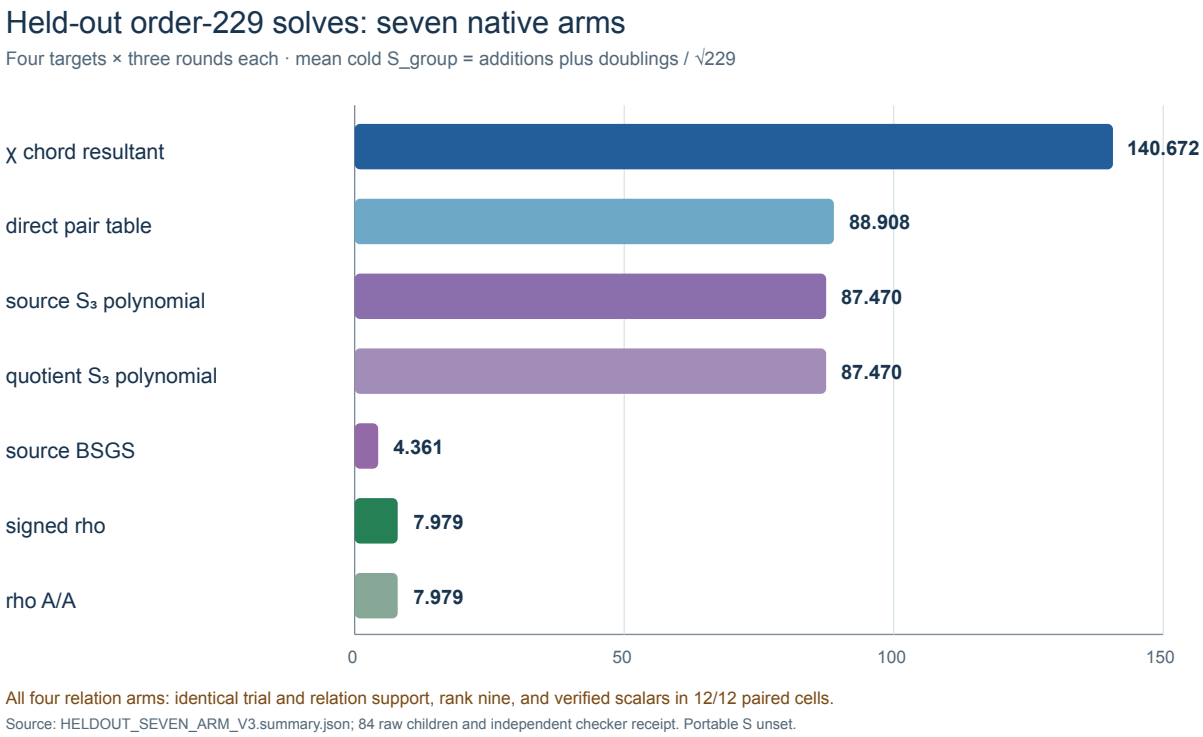
S3 decomposition and seven-arm held-out solves

Run: heldout-seven-arm-v3 · **field:** `F_211^2` · **subgroup:** order 229 · **host:** Darwin arm64 shared L0.



Both S3 polynomial oracles matched independent pair enumeration on all 229 subgroup points: 136 pairs and 130 zero-pair cases per curve. All 84 cold children returned scalars independently replayed on both different-j curves. The checker replayed all 664 relation witnesses. The four relation arms had identical trials, relation counts and rank-nine outcomes in all 12 paired round-target cells.

Mean cold counted-group S: χ 140.672, direct pair 88.908, source and quotient S3 87.470 each, signed rho and A/A 7.979 each, and BSGS 4.361. Source S3/rho = 10.962 counted-group operations; χ/rho = 17.629. Mean whole-child CPU was 5.824 ms for source S3 and 5.791 ms for rho; rho A/A/rho = 1.058.

The group count omits unconverted field and allocation costs; field counters overlap group arithmetic and miss some square-root internals. Shared-host CPU is descriptive; complete portable S and isolated wall admission remain open. Direct inverse-Vélu and full-image relation arms on this cell remain open. Sources: `S3_EXHAUSTIVE_SPEC_V2.json`, `HELDOUT_SEVEN_ARM_SPEC_V3.json`, `HELDOUT_SEVEN_ARM_V3.summary.json`, independent receipts, and 84 raw children.